

Integrated Rate Laws and Reaction Mechanisms

Answer key with worked solutions

A. Understanding

1. Definitions.

(a) **Half-life:** the time for the concentration of a reactant to fall to half its value. (b) **Elementary step:** a single step in a mechanism, whose rate law follows from the particles that collide. (c) **Rate-determining step:** the **slowest** step, which sets the overall rate. (d) **Intermediate:** a species **made** in one step and **used up** in a later one.

2. Plots and units.

Zero: [A] against t, slope -k, k in M s^{-1} . **First:** $\ln[A]$ against t, slope -k, k in s^{-1} . **Second:** $1/[A]$ against t, slope +k, k in $\text{M}^{-1} \text{s}^{-1}$.

B. Calculations

3. Drug elimination.

$$\begin{aligned} \text{(a)} \quad k &= \frac{\ln 2}{t_{1/2}} = \frac{0.6931}{4.00 \text{ h}} = \mathbf{0.173 \text{ h}^{-1}} \\ \text{(b)} \quad n &= \frac{10.0 \text{ h}}{4.00 \text{ h}} = 2.50 \text{ half-lives}; 200. \text{ mg} \times \left(\frac{1}{2}\right)^{2.50} = \mathbf{35.4 \text{ mg}} \end{aligned}$$

4. First-order time.

$$t = \frac{\ln([A]_0/[A])}{k} = \frac{\ln(0.800 \text{ M}/0.200 \text{ M})}{0.0200 \text{ min}^{-1}} = \frac{1.386}{0.0200 \text{ min}^{-1}} = \mathbf{69.3 \text{ min}}$$
 (exactly two half-lives of 34.7 min)

5. Second order.

$$\begin{aligned} \text{(a)} \quad t_{1/2} &= \frac{1}{(0.150 \text{ M}^{-1} \text{s}^{-1})(0.0500 \text{ M})} = \mathbf{133 \text{ s}} \\ \text{(b)} \quad \frac{1}{[A]} &= \frac{1}{0.0500 \text{ M}} + (0.150 \text{ M}^{-1} \text{s}^{-1})(200. \text{ s}) = 20.0 \text{ M}^{-1} + 30.0 \text{ M}^{-1} = 50.0 \text{ M}^{-1}, \text{ so } [A] = \mathbf{0.0200 \text{ M}} \end{aligned}$$

6. Zero order.

$$(a) t_{1/2} = \frac{0.500 \text{ M}}{2(2.50 \times 10^{-3} \text{ M/s})} = \mathbf{100. \text{ s}} \quad (b) t = \frac{0.500 \text{ M}}{2.50 \times 10^{-3} \text{ M/s}} = \mathbf{200. \text{ s}}$$

C. Order from data

7. First order from data.

In each 100 s interval [A] is multiplied by the **same fraction**: $\frac{0.300}{0.400} = 0.750$, $\frac{0.225}{0.300} = 0.750$, $\frac{0.169}{0.225} = 0.751$. A constant fraction in equal times means **first order** ($\ln[A]$ against t is a straight line).

$$k = \frac{\ln(0.400 \text{ M}/0.300 \text{ M})}{100 \text{ s}} = \frac{0.2877}{100 \text{ s}} = \mathbf{2.88 \times 10^{-3} \text{ s}^{-1}}$$

D. Mechanisms

8. Mechanism.

(a) Add the steps and cancel Cl: $2 \text{ NO}_2\text{Cl} \longrightarrow 2 \text{ NO}_2 + \text{Cl}_2$ (b) **Cl** (made in step 1, used in step 2) (c) The slow step has one NO_2Cl : **rate** = $k[\text{NO}_2\text{Cl}]$

E. Challenge

9. Successive half-lives.

$$\text{First: } t_{1/2} = \frac{1}{(0.0500 \text{ M}^{-1} \text{ s}^{-1})(0.100 \text{ M})} = \mathbf{200. \text{ s}}. \text{ Second: } t_{1/2} = \frac{1}{(0.0500 \text{ M}^{-1} \text{ s}^{-1})(0.0500 \text{ M})} = \mathbf{400. \text{ s}}$$

Each half-life is **twice** the one before. If measured half-lives keep doubling, the reaction is second order (first order gives equal half-lives).